

LiftMetrics

Video analysis of bar path and bar velocity
for Olympic lifting and strength training

Snatch · Clean · Clean & jerk

Back squat · Front squat · Deadlift · Bench press

A lift · A complex · A session

Presentation and ambassador proposal

Prepared for Teti Xanthoudaki

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1 Introduction

In Olympic lifting, the difference between a made lift and a miss often comes down to a few centimetres: a bar that swings forward in the second pull, a catch too far in front, a drop under the bar that takes too long. These faults are hard to see with the naked eye, in real time, at the speed of a snatch.

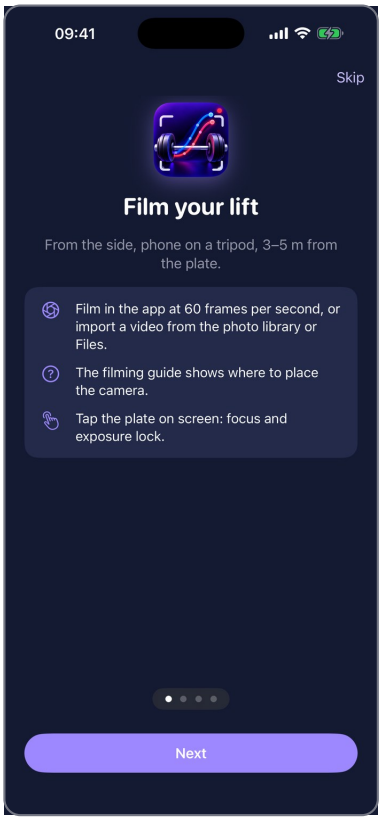
LiftMetrics is an iPhone app that **films the lift from the side, tracks the plate frame by frame and draws the bar path**. It splits the movement into phases, measures what a coach needs (horizontal displacements, heights, velocities, timings), gives a **technique score out of 100** and offers **concrete coaching**: cues and drills for each fault.

In short

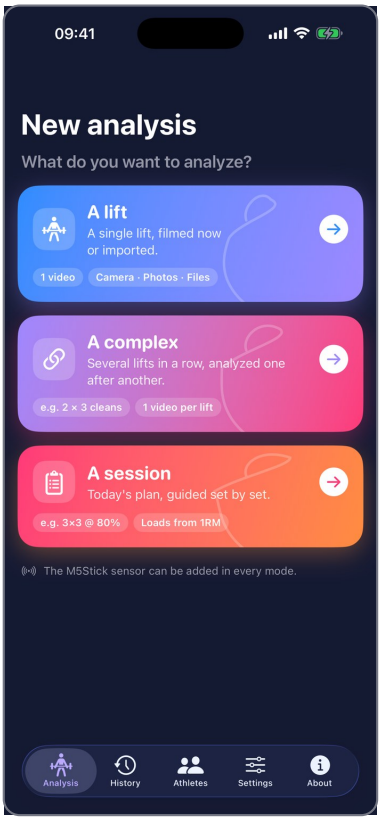
- An iPhone on a tripod is all you need: the side-view video is the only requirement.
- An optional M5StickS3 sensor on the bar sleeve makes velocity more accurate and measures bar tilt.
- Results in under a minute: bar path coloured by phase, curves, score, coaching cues.
- Squat, deadlift and bench press: film the whole set at once and get the velocity of every rep plus velocity loss (velocity-based training).
- Comparison with the world's best lifters (2018 IWF World Championships) and a national-level team.
- PDF reports, images and videos for Instagram, and a file format to share lifts between users.
- Everything is computed and stored on the iPhone: no account, no video uploaded.

Who is it for?

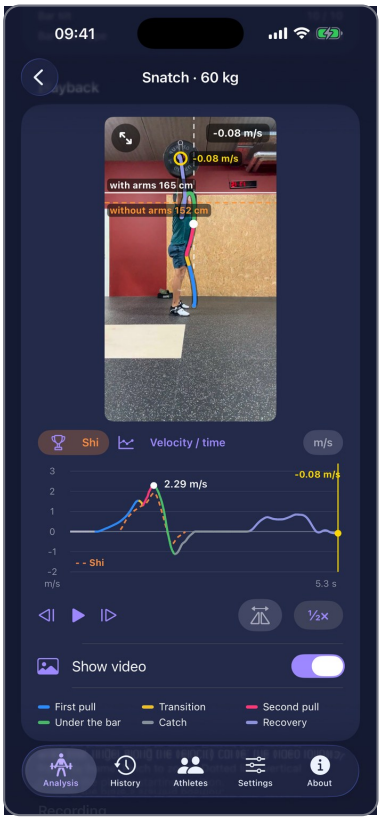
- **Athletes**, to understand their lifts and track progress session after session.
- **Coaches**, to back up what they see with numbers, compare an athlete's lifts and build programmes.
- **Boxes, clubs and events**, to offer instant analysis to athletes and produce visuals to share.



Welcome screen (first launch)



Choosing the type of analysis



Replay of an analysed lift

2 Why it matters for CrossFit athletes

Barbell cycling, heavy singles and strength pieces are at the heart of CrossFit training and competition. **LiftMetrics** brings lab-style barbell feedback to any box, with nothing more than a phone:

- **Snatch and clean & jerk technique** under fatigue or at heavy loads: see exactly where the bar drifts, how deep and how fast the athlete drops under it, and whether the catch position holds.
- **Velocity-based strength work** on back squat, front squat, deadlift and bench press: mean and peak velocity of every rep and velocity loss across the set — a simple, objective way to manage fatigue during a training block.
- **Complexes and sessions**: chain several lifts (for example 1 clean + 1 front squat + 1 jerk) or follow the day's programme set by set, with loads computed from the athlete's 1RMs.
- **No extra hardware needed**: side-view video is enough. The sensor is optional, for those who want more precision.
- **Content ready to post**: an Instagram image or slow-motion video of a lift with its bar path and live velocity.

Typical use

The athlete films a heavy snatch double on the iPhone, sees the bar path and the score within a minute, then sends the lift file to the coach, who opens it in their own app with all the curves and measurements.

3 System and environment

Main device	iPhone running iOS 17 or later (iPhone app only, no iPad).
Camera	iPhone rear camera, 60 fps in HD 1080p (4K works but adds no useful precision).
Sensor (optional)	M5Stack **M5StickS3** (BMI270 accelerometer and gyroscope) running the "BarPathLogger" firmware; the M5StickC Plus2 is also supported.
Connections	Bluetooth for the sensor. No internet connection needed.
Languages	English, French, Spanish, German, Italian (chosen in Settings).
Data	Local database on the iPhone; videos in Photos or in Files → LiftMetrics. No server, no account.

Why the iPhone?

The iPhone camera natively films at 60 frames per second with a very steady frame rate, which makes velocity calculations reliable. All computations (plate tracking, phases, measurements) run on the device: analysis works in a gym with no network.

4 Required and optional hardware

Hardware	Status	Role
iPhone with LiftMetrics	**required**	Films (or imports) the side-view video, runs the analysis, keeps the history.
Tripod or stable stand	**required**	The camera must not move: any shake distorts the bar path.
Plates of known diameter	**required**	The plate facing the camera sets the scale (45 cm by default, adjustable).
M5StickS3 + sleeve mount	optional	More accurate velocity (200 samples/s), bar tilt and rotation, power.

4.1 Two possible setups

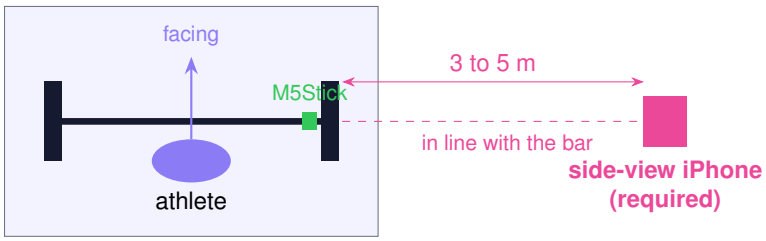
Setup	What you need	Best for
A — iPhone only	1 iPhone + tripod	Getting started, gyms with no equipment, importing existing videos.
B — iPhone + M5Stick	1 iPhone + tripod + sensor on the sleeve	Accurate tracking of velocity, power and bar tilt.

4.2 The M5StickS3 sensor

- Records the accelerometer (±16 g) and gyroscope (±2000 °/s) at **200 Hz** in its memory.
- Controlled by the app over Bluetooth: starts and stops with the camera, clock sync, file transfer then erase.
- Button A: start or stop recording (it can also trigger the iPhone: see §7).
- The screen shows whether it is recording on its own or on the iPhone's request.
- Fallback mode: Wi-Fi access point with a page to download the files.

5 Setup

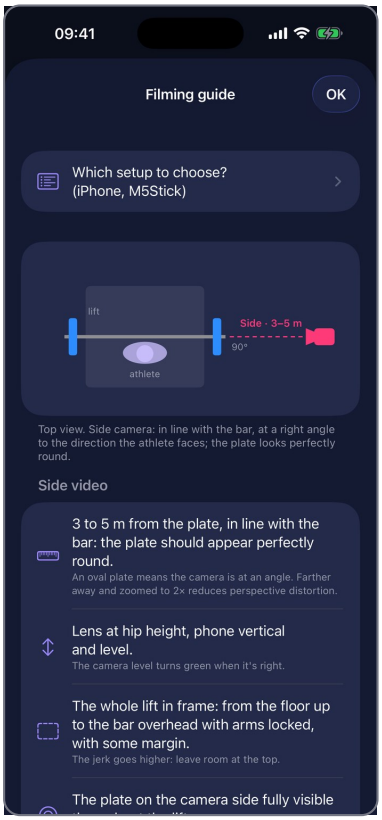
5.1 Positioning the camera



platform (top view)

The built-in filming guide recalls the key rules:

- Side view in line with the bar, 3–5 m away: the plate must look **perfectly round** (oval = camera at an angle).
- Lens at hip height, phone upright and level (the on-screen level turns green).
- The whole movement in frame, from the floor to arms locked overhead (the jerk goes higher).
- Plate facing the camera fully visible, nothing in front of it.
- 60 fps; tapping the plate on screen locks focus and exposure.
- Good light, avoid backlighting.
- Start 2 s before lift-off, stop 2 s after the catch: the still bar serves as a reference.

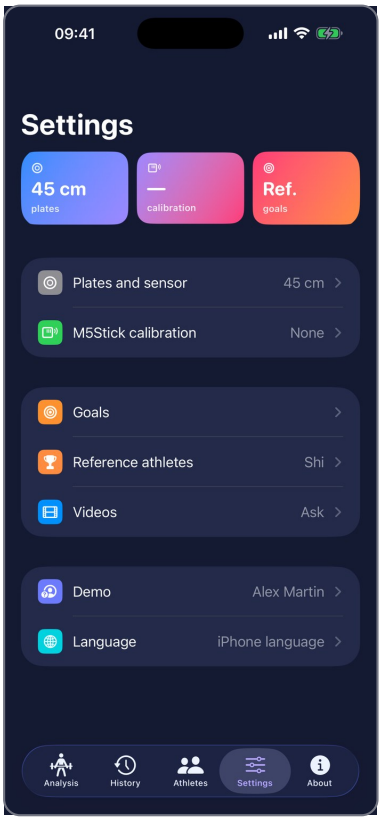


Filming guide

5.2 Settings

Settings are organised like on an iPhone, one page per topic:

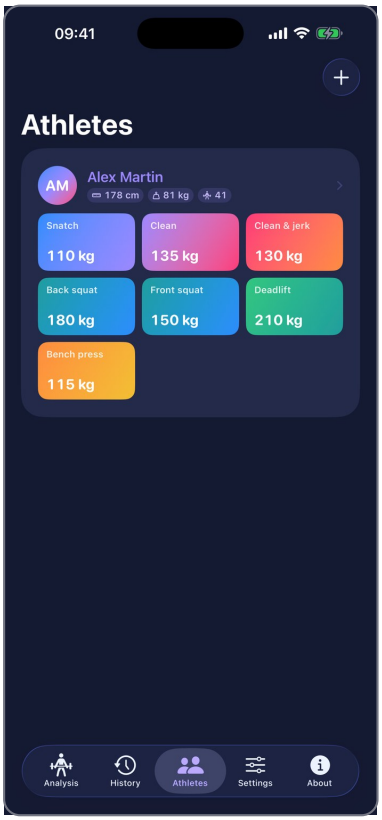
- **Plates and sensor**: plate diameter (scale), sleeve side where the sensor is mounted.
- **M5Stick calibration**: guided 6-position wizard (one per face), with audio cues.
- **Goals**: target ranges for each measurement, per movement (snatch, clean, jerk).
- **Reference athletes**: athlete shown on the charts and alignment mode (§12).
- **Videos**: after a lift is saved, delete full-length videos, ask, or keep everything (the app always keeps a trimmed clip of the lift).
- **Language**: the iPhone's language or one of the app's 5 languages.



Settings

5.3 Athletes

Each athlete has a profile: **height** (heights shown as % of height), **body mass** (power per kg), **category** women/men (automatic choice of the reference athlete) and **dated 1RMs** per movement. The percentage of max for a lift uses the 1RM valid on its date; a session programme computes loads from these 1RMs.



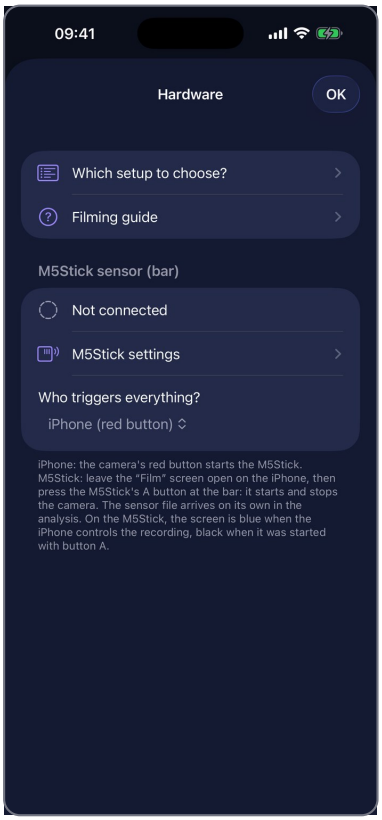
Athlete profile and 1RMs

5.4 Connected hardware

The **Hardware** screen connects the M5Stick, gives access to its settings and defines **who triggers everything**:

- **the iPhone**: the camera's red button starts the video and the sensor;
- **the M5Stick**: one press on the sensor's button A, mounted on the bar, starts and stops everything — handy when the athlete trains alone.

The sensor's battery level shows on the iPhone, and its file flows into the analysis automatically.



Hardware

6 Installation

1. **LiftMetrics** on the main iPhone: App Store (release planned) or TestFlight during the beta.
2. On first launch, a 4-page introduction explains the concept; it remains available in About.
3. **Permissions** requested when needed: camera, Photos (importing and saving videos), Bluetooth (sensor).
4. **Sensor option**: M5StickS3 with the BarPathLogger firmware (M5Unified library, "M5StickS3" board in the Arduino IDE). The sensor can be supplied pre-programmed. Connect it in Hardware, then run the guided calibration once.
5. Create the athlete profile (height, body mass, category, 1RMs).

Setup time

Setup A: 2 minutes (put the iPhone on a tripod). Setup B: 5 minutes the first time (pairing and calibration), then the app finds the hardware on its own.

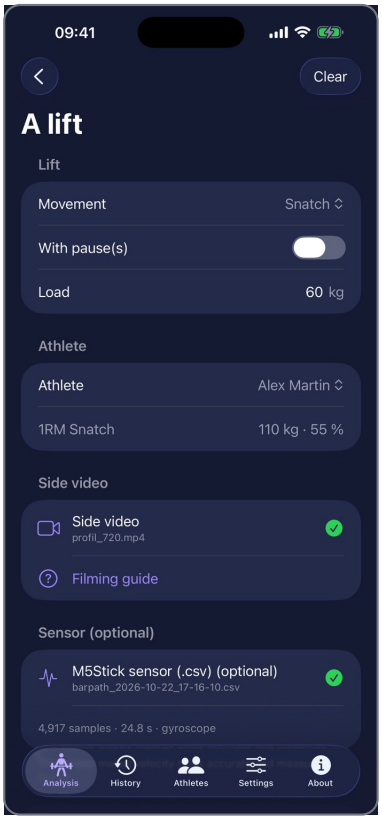
7 Modes

The home page of the **Analysis** tab offers three modes. Once a mode is chosen, the page only shows what is relevant to it; the "‹" arrow goes back to the choice.

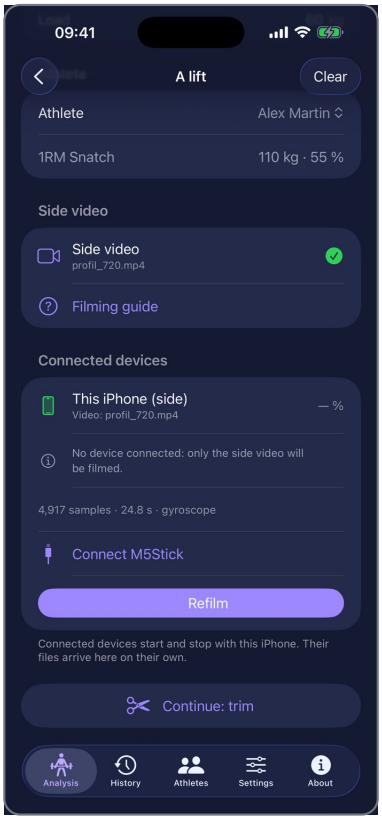
Mode	Principle	Example
A lift	One lift, filmed now or imported (Photos, Files).	1 snatch at 80 kg.
A complex	Several lifts chained, one video per lift; they are analysed one after another and grouped in the history.	2 sets of 3 cleans; 1 snatch + 1 clean & jerk.
A session	The day's programme, guided set by set; loads computed from 1RMs.	3×3 cleans @ 80 % + 2×1 clean & jerk @ 70 %.

7.1 A lift

Choose the movement (snatch, clean, clean & jerk, with or without pause; back squat, front squat, deadlift, bench press), the load and the athlete, then the side-view video: **film** with the connected devices, or **import** from Photos or Files (a sensor file can then be added).



A lift: movement, load, athlete, video



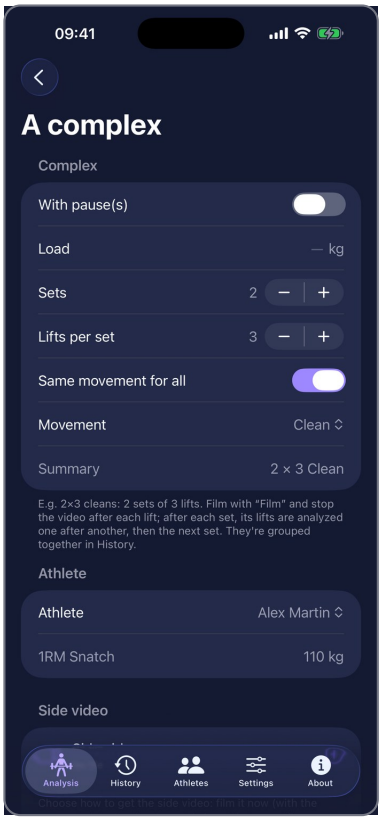
Connected devices and camera

7.2 A complex

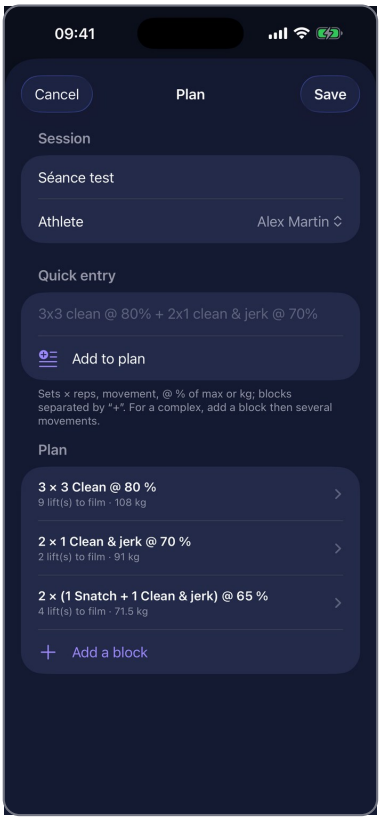
Number of sets, number of lifts per set and the movement of each lift (the same for all, or different). Stop the video after each lift; the sensor keeps its files until the end of the set, then the lifts are analysed one after another before the next set.

7.3 A session (programme)

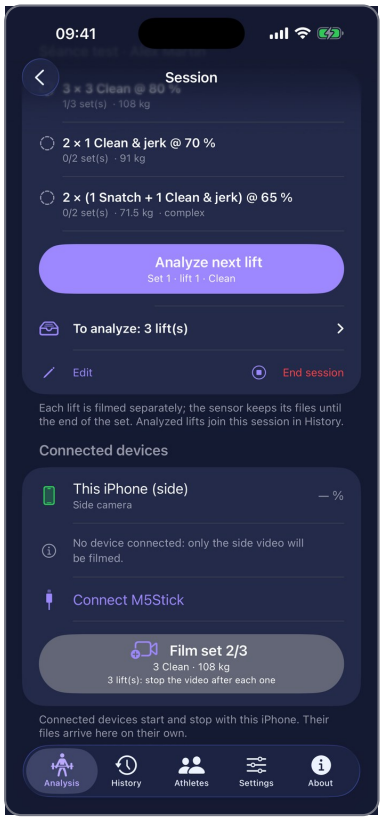
The programme is typed in plain text (“3x3 clean @ 80% + 2x1 clean & jerk @ 70%”) or block by block; a block can be a complex. The app shows the next set and its computed load, lets you film the set, then analyse each lift in turn. Analysed lifts join the session in the history.



A complex



Preparing the programme

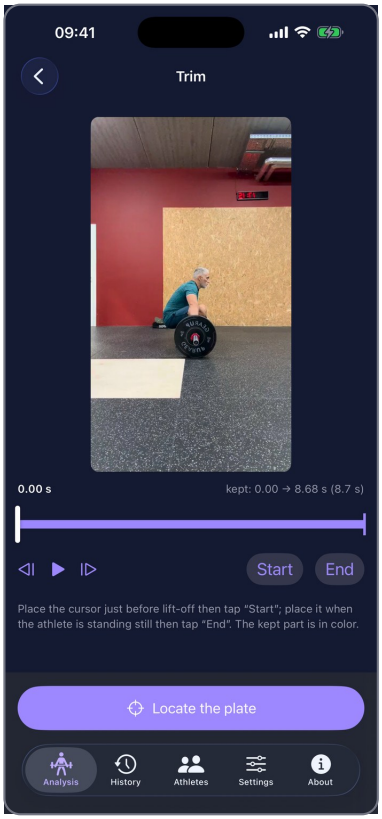


Session: analyse, then film the next set

8 Analysing a lift

8.1 Steps

1. **Video** filmed or imported (with the sensor file if any).
2. **Trim**: keep a few seconds around the lift.
3. **Plate detection**: one tap on the plate; adjust the circle if needed (pinch to zoom).
4. **Analysis**: frame-by-frame plate tracking, scaling from the plate diameter, automatic sync with the sensor, phase detection, measurements.
5. **Results**: score, coaching, replay, measurements (under a minute on a recent iPhone).



Trimming around the lift

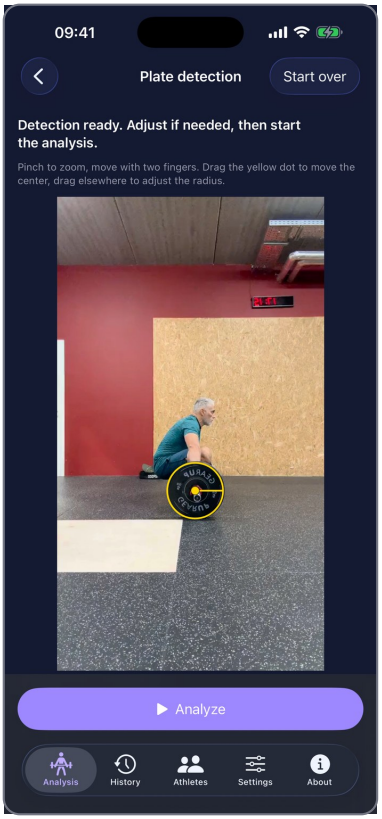
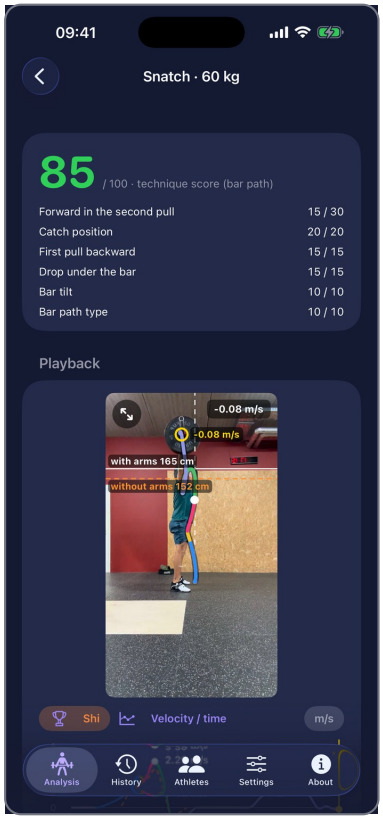


Plate detection



Score out of 100 and details

8.2 Phases of the movement

The bar path is coloured by phase, the same way on the video and on the curves:

Phase	Boundaries
First pull	from lift-off to the transition (end of knee extension)
Transition	knees re-bend under the bar
Second pull	from the transition to peak velocity
Under the bar	from peak velocity to the catch
Catch	stabilisation, then recovery to standing
Jerk	dip, drive, lockout

8.3 Checks and corrections

- **Lifter direction** detected automatically (facing left or right), editable.
- **Tracking correction**: if the plate is briefly hidden, it can be repositioned and tracking restarted from that point.
- **Video–sensor sync** can be checked and adjusted to the hundredth of a second.
- **Re-analysis** of a lift from the history using its saved videos and files.

9 Measurements by hardware

All horizontal measurements are relative to the vertical line of the bar's starting position; heights are measured from the floor (plate centre) and as % of the athlete's height when known.

Measurement	iPhone only	+ M5Stick
Bar path		
Bar path coloured by phase	✓	✓
Bar sweep in the first pull (bar pulled back toward the athlete)	✓	✓
Forward displacement in the second pull	✓	✓
Loop and catch position	✓	✓
Bar path type (1, 2 or 3)	✓	✓
Peak height, catch height, drop under the bar	✓	✓
Height without arm pull, arm contribution	✓	●
Velocity and timing		
Peak velocity, velocity at the end of the first pull	✓	●
Mean velocity, first and second pull	✓	●
Peak acceleration, first and second pull	✓	●
Durations: first pull, second pull, turnover	✓	●
Catch stabilisation, recovery, pauses	✓	✓
Peak power (W and W/kg)	—	✓
Bar orientation		

Measurement	iPhone only	+ M5Stick
Peak tilt (one side higher)	—	✓
Peak horizontal rotation (one side ahead of the other)	—	✓
Jerk		
Dip depth and duration	✓	●
Peak drive velocity	✓	●
Displacement during the drive, lockout position	✓	✓

✓ measured ● measured more accurately with the sensor (200 samples/s, orientation tracked by the gyroscope) — not available

Video velocity vs sensor velocity

Without a sensor, velocity is computed from the plate position 60 times per second. With the M5Stick, acceleration measured 200 times per second is fused with the video: the curve is finer, especially at peak velocity and during the turnover. The moments when the bar is still, before and after the lift, are used to re-anchor the velocity.

10 Strength lifts: squat, deadlift, bench press

LiftMetrics also tracks four strength lifts: **back squat, front squat, deadlift and bench press**. They are picked from the same “Movement” list as the Olympic lifts, with no extra menu, and are available everywhere: athlete 1RMs, session programmes (“5x5 back squat @ 80%”), history filters.

10.1 One video for the whole set

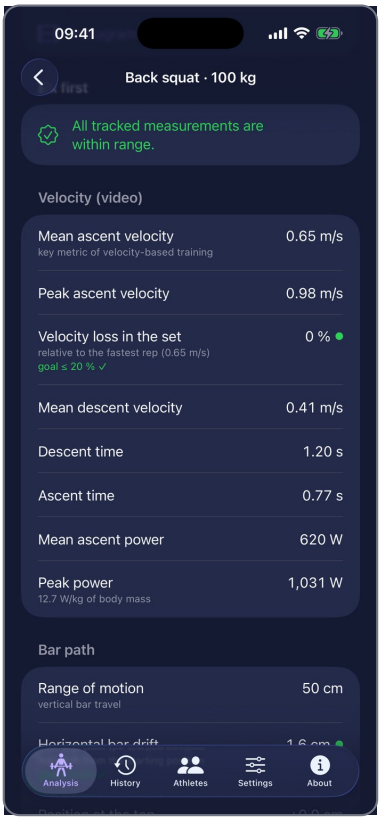
Unlike Olympic lifts (one video per lift), you film **the whole set in a single video**. The app detects each rep and analyses it separately: a set of 5 squats gives **5 analyses**, saved in one go and grouped in the history under the **SET** label, like a complex. Each rep keeps its own trimmed clip.

Movement	How a rep is detected
Squats, bench press	descent (at least 15 cm for the squat, 10 cm for the bench), bottom position, ascent until the bar stops at the top
Deadlift	lift-off from the floor, rise of at least 25 cm, lockout, return to the floor

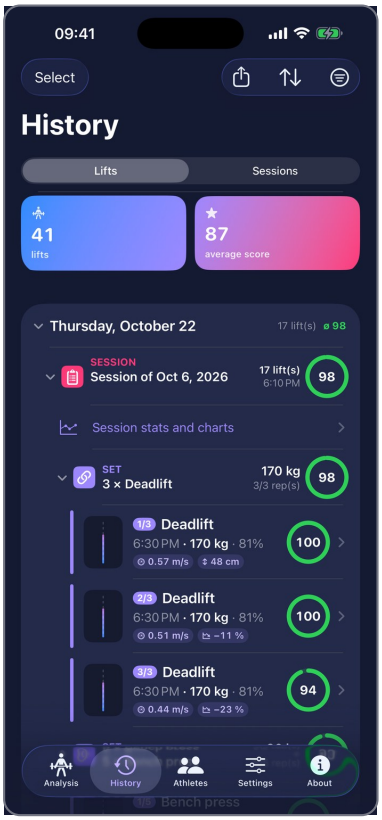
Phases are coloured as for Olympic lifts: **descent**, **ascent**, **sticking point** (if velocity dips during the ascent), **top**.



Set of 5 squats: velocity of each rep



Measurements for one rep



Set grouped in the history

Screenshots taken with a synthetic test video (5 squats, each slower than the last).

10.2 Measurements for each rep

Measurement	Definition	Hardware
Mean ascent velocity	vertical displacement / duration of the ascent: the reference measure of velocity-based training	iPhone
Peak ascent velocity	velocity peak during the ascent	iPhone
Velocity loss in the set	drop from the fastest rep (in %)	iPhone
Descent velocity and duration, ascent duration	tempo of the rep	iPhone
Sticking point	velocity and height (% of range) of the velocity dip during the ascent	iPhone
Range of motion	vertical travel of the bar	iPhone
Horizontal deviation	maximum bar deviation from its starting position	iPhone
Top position	final position relative to the start	iPhone
Mean and peak power	from the entered load	iPhone (+ sensor)
Bar tilt and rotation	one side higher, or ahead of the other	M5Stick

10.3 Score and goals

Criterion	Points	Default goal
Vertical bar path (horizontal deviation)	45	$\leq$ 5 cm squat, $\leq$ 4 cm deadlift, $\leq$ 10 cm bench press
Velocity loss in the set	35	$\leq$ 20 %
Bar tilt (sensor)	20	$\leq$ 3°

On the bench press the bar naturally follows a J-curve, so the horizontal-deviation goal is wider. Velocities and range of motion depend on the load: they are not tracked by default but can be (for example a target mean-velocity zone for power work).

Why velocity loss?

In velocity-based training, the drop in mean velocity across a set reflects fatigue. Stopping the set once the loss exceeds a threshold (often 20 %) lets you dose the effort without going to failure. The app shows this loss rep by rep, in orange beyond the goal.

Filming a set

Side view, plate facing the camera fully visible, as for Olympic lifts. For squat and deadlift, lens at hip height; for bench press, camera at bench height. Start before unracking the bar (or before the first lift-off) and stop after the last rep. Comparison with reference athletes is not yet available for these lifts.

11 Reports

11.1 For each lift

Technique score out of 100. It summarises the bar path against the goals you set:

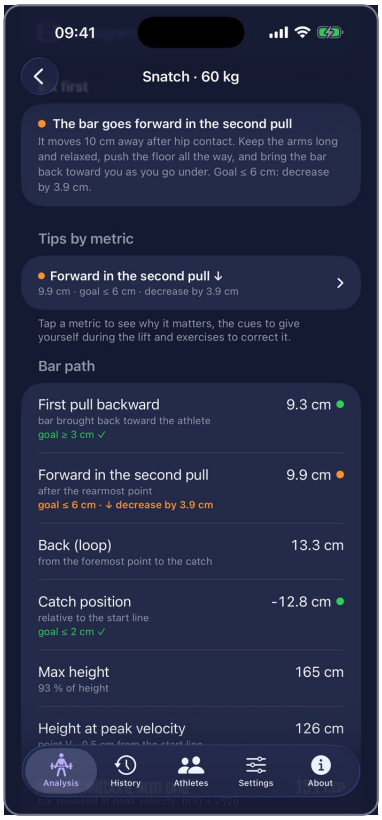
Criterion	Points
Forward in the second pull	30
Catch position	20
First pull backward	15
Drop under the bar	15
Bar tilt (sensor)	10
Bar path type	10
Clean & jerk: the jerk counts for 30 % of the score	

Each tracked measurement is **green** inside the range, **orange** slightly outside, **red** beyond. The app tells you whether to increase or decrease it, and by how much.

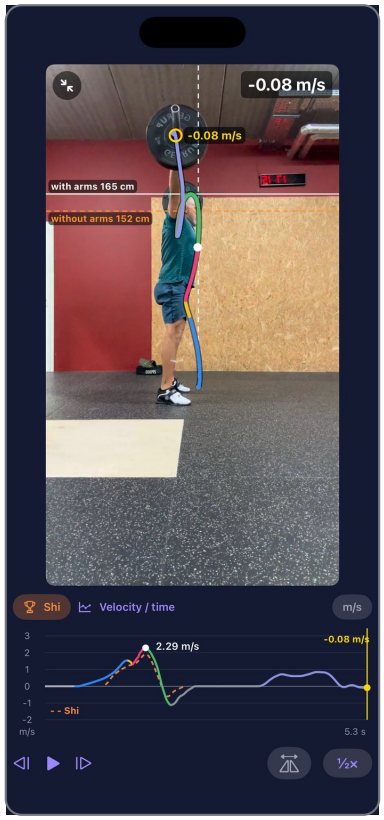
Coaching. The top three priorities, then for each measurement outside its goal: why it matters, cues to give during the lift and drills to fix it.

Replay. The video replays with the bar path drawing itself and the velocity shown at every instant; scrub through the lift by touching the curve. Slow motion ½ and ¼, frame by frame, mirror. The double arrow on the video switches to **full screen**.

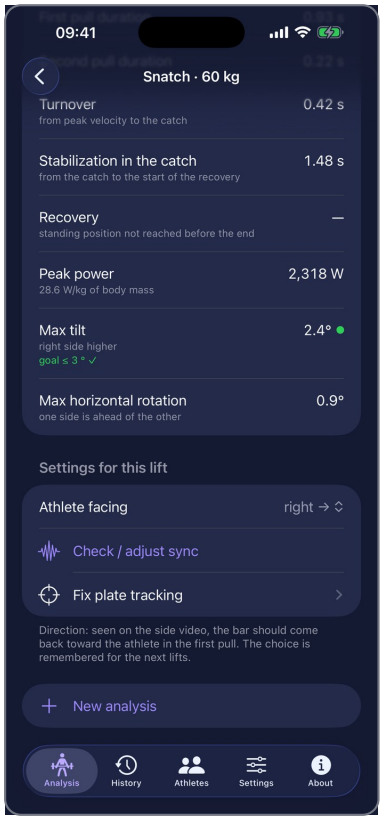
Charts. Four curves to choose from: velocity vs time, velocity vs height, acceleration vs time, acceleration vs height. Velocity can be shown in m/s or as % of peak velocity.



Priorities and measurements



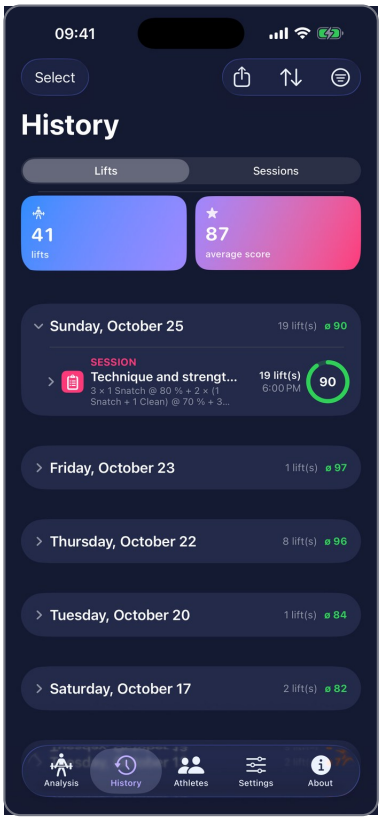
Full-screen replay



Velocity and timing measurements

11.2 History, sessions and tracking

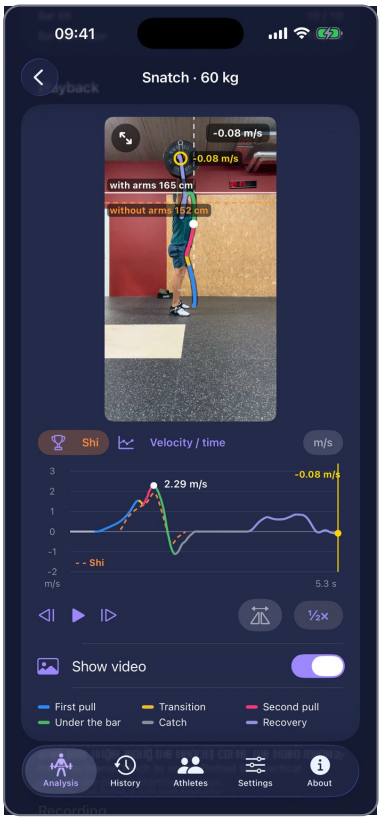
- **History**: every lift, grouped by day, with a bar-path thumbnail, the score and the key measurements; lifts of a complex are grouped and labelled. Sorting, filters (movement, athlete, make/miss), selection of lifts to analyse a session.
- **Sessions**: session statistics, average score, table of lifts and comparison of measurements.
- **Personal goals**: “From my lifts” turns the best quarter of made lifts into progression goals.
- **History exports** to PDF (table of lifts).



History (grouped complex)



Session



Lift compared with Shi Zhiyong

12 Comparison with elite lifters

The app includes **reference snatches** digitised from scientific publications:

- **World elite** — 2018 IWF World Championships (Ashgabat): the best lifter in each category, **10 women and 9 men** (Om Yun Chol, Shi Zhiyong, Sohrab Moradi, Lasha Talakhadze, Sukanya Srisurat, Tatiana Kashirina. . .), with velocity curve, horizontal velocity and bar path.
- **National level** — Singapore national team (3 men, 3 women, at 80–90 % of 1RM): bar paths and reference values, closer to a good club level.

On the charts, the chosen athlete's velocity curve appears as a dashed orange line. The choice can be:

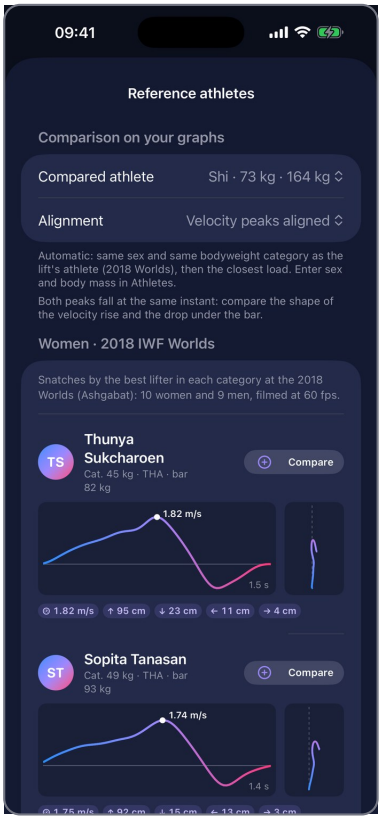
- **automatic**: same sex and bodyweight category as the athlete of the lift, then the closest load;
- **manual**: any athlete in the list.

Time alignment, three options:

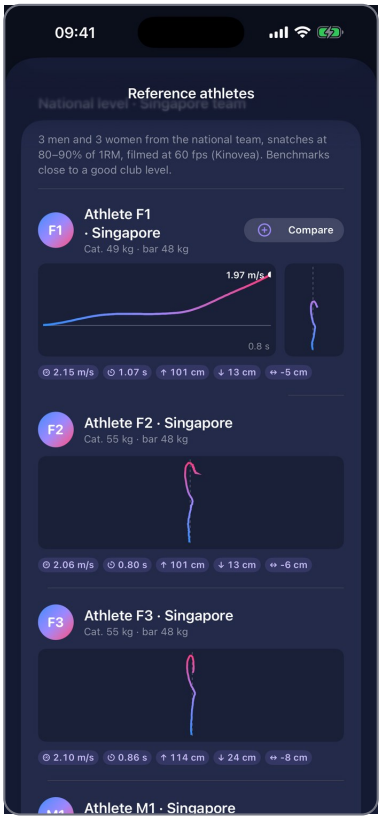
Peaks aligned	Both velocity peaks happen at the same instant: compare the shape of the acceleration and the turnover.
Lift-off and peak aligned	The reference curve is stretched or compressed: compare phases regardless of pull duration.
Lift-offs aligned	Real time: see who reaches peak velocity faster.

Clean and jerk. No athlete-by-athlete curves have been published; the app gives numerical bench-

marks: peak clean height ≈ 59.5 % of height (men) and 62–68 % (women), drop under the bar 26 to 42 cm; in the jerk, dip ≈ 18 cm and drive velocity 1.53 m/s among elite lifters.



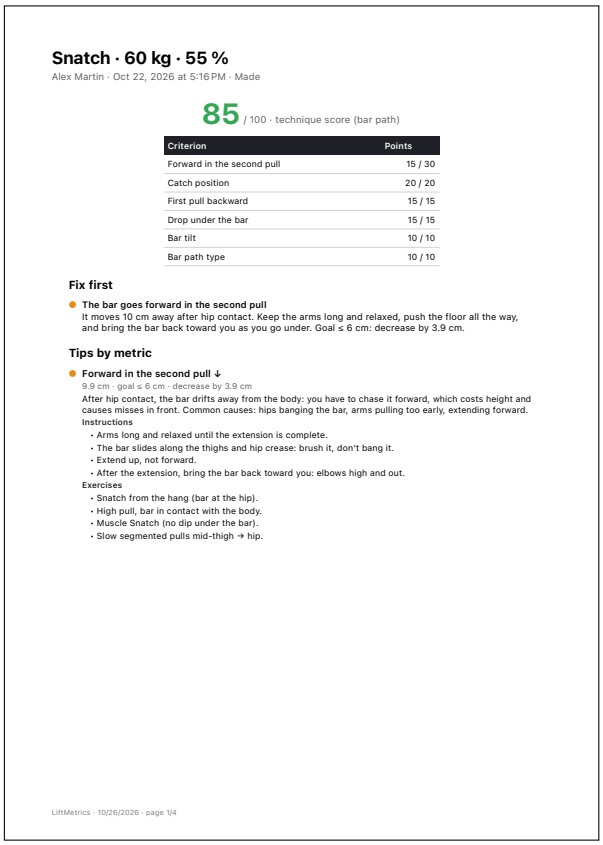
Reference athletes (Settings)



National level (Singapore)

13 PDF report sample

Each lift exports to PDF: header (movement, load, % of max, athlete, date), score and breakdown, priorities with cues and drills, then bar path, curves and the full measurement table on the following pages. History and sessions also export as tables.



Snatch · 60 kg · 55 %

Alex Martin · Oct 22, 2026 at 5:16 PM · Made

85 / 100 · technique score (bar path)

Criterion	Points
Forward in the second pull	15 / 30
Catch position	20 / 20
First pull backward	15 / 15
Drop under the bar	15 / 15
Bar tilt	10 / 10
Bar path type	10 / 10

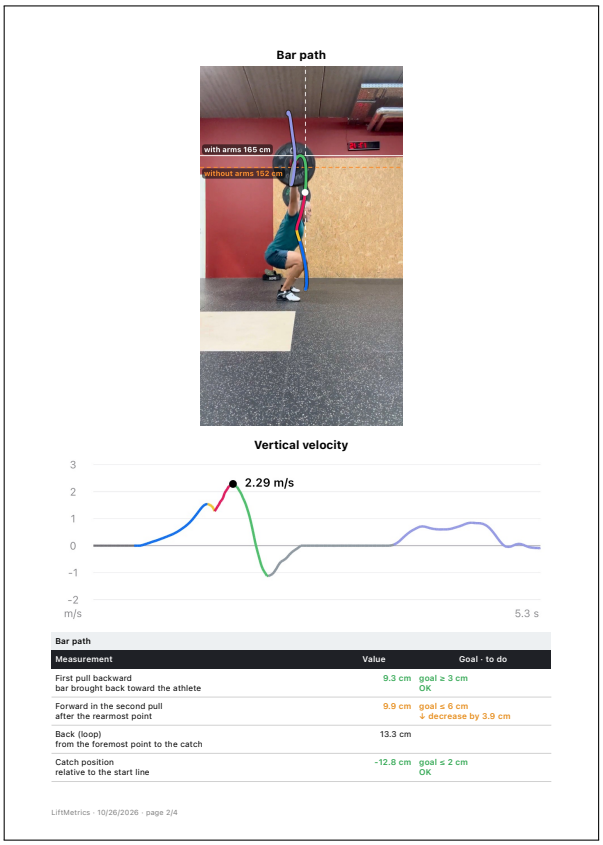
Fix first

- **The bar goes forward in the second pull**
  It moves 10 cm away after hip contact. Keep the arms long and relaxed, push the floor all the way, and bring the bar back toward you as you go under. Goal ≤ 6 cm: decrease by 3.9 cm.

Tips by metric

- **Forward in the second pull ↓**
  9.9 cm · goal ≤ 6 cm · decrease by 3.9 cm
  After hip contact, the bar drifts away from the body: you have to chase it forward, which costs height and causes misses in front. Common causes: hips banging the bar, arms pulling too early, extending forward.
  Instructions
  - Arms long and relaxed until the extension is complete.
  - The bar slides along the thighs and hip crease: brush it, don't bang it.
  - Extend up, not forward.
  - After the extension, bring the bar back toward you: elbows high and out.
  Exercises
  - Snatch from the hang (bar at the hip).
  - High pull, bar in contact with the body.
  - Muscle Snatch (no dip under the bar).
  - Slow segmented pulls mid-thigh → hip.

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Bar path

with arms 165 cm

without arms 152 cm

Vertical velocity

3
2
1
0
-1
-2
m/s

2.29 m/s

5.3 s

Bar path

Measurement	Value	Goal · to do
First pull backward bar brought back toward the athlete	9.3 cm	goal ≥ 3 cm OK
Forward in the second pull after the rearmost point	9.9 cm	goal ≤ 6 cm ↓ decrease by 3.9 cm
Back (loop) from the foremost point to the catch	13.3 cm	
Catch position relative to the start line	-12.8 cm	goal ≤ 2 cm OK

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Pages 1 and 2 of the PDF report for a 60 kg snatch

14 Instagram sample

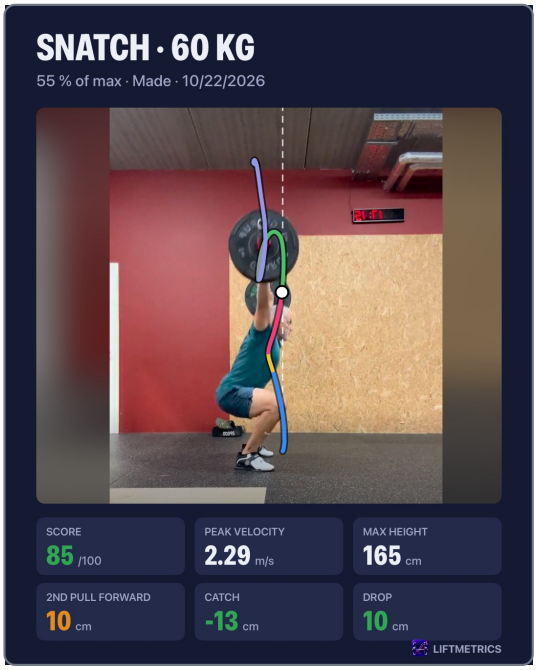
Two ready-to-post formats:

- **Image** (1080×1350): photo of the catch with the bar path, score and 5 key measurements.
- **Video**: the lift in slow motion, with the bar path drawing itself and live velocity, choice of playback speed and on-screen information.

Measurement colours follow the goals (green, orange, red). The LiftMetrics logo sits at the bottom of the image.

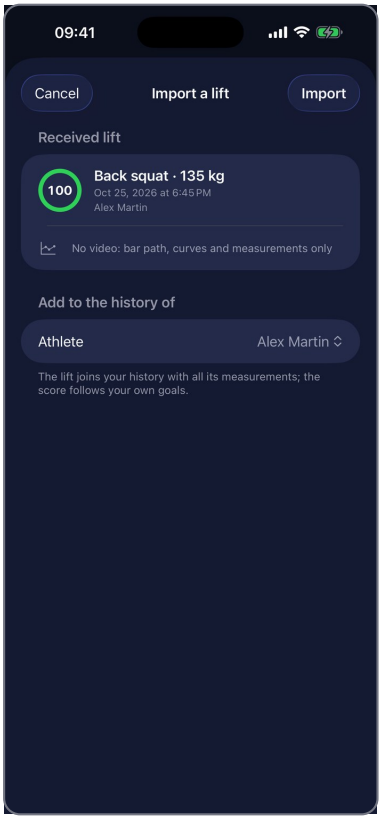
Content idea

A weekly “bar path of the week”: one heavy lift, its path and velocity, and the one cue that made the difference.



15 Sharing a lift

LiftMetrics file (.liftmetrics). An analysed lift can be sent via AirDrop, Messages, Mail or WhatsApp, with or without its trimmed clip. The recipient taps it: LiftMetrics opens, shows a preview and files the lift in their history, under the athlete of their choice. They get all the measurements, curves and replay; the score follows their own goals.
CSV. Curves (time, position, velocity) export for a spreadsheet or scientific processing.
Typical use: the athlete films at home or at the box, the coach receives the lift and reviews it with the same tools.



Importing a received lift

16 At competitions and events

LiftMetrics can support a competition or a training camp without disrupting it:

- **An analysis station beside the platform**: one iPhone on a tripod, in line with the bar, 3–5 m away. Lifts are filmed and analysed during the event; each athlete leaves with their report.
- **A warm-up station**: instant feedback between attempts (bar path, velocity, one thing to fix), useful for coaches.
- **Visuals for the event**: Instagram images and videos of the best lifts, comparison with world references.
- **One file per athlete** (.liftmetrics or PDF) handed over after the event, which the athlete can open in their own app.
- **Strength events**: the same station works for squat, bench press and deadlift (velocity and bar path of every attempt).

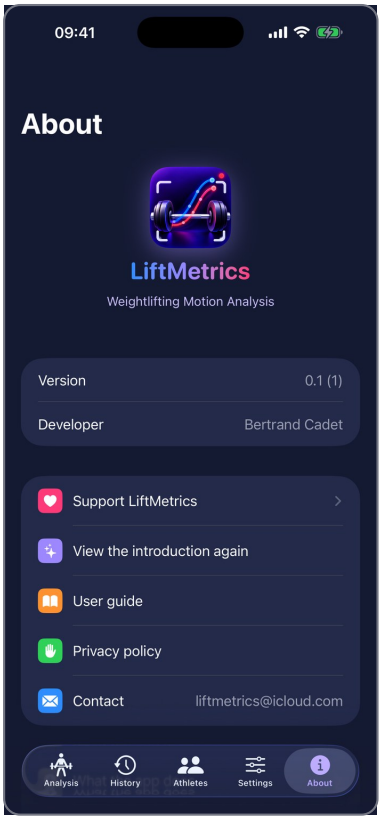
Practical points

- Free space beside the platform, in line with the bar, with nobody walking in front of the camera.
- Enough light on the platform; no backlighting behind the athlete.
- Athletes' consent to be filmed and for images to be shared.
- No data leaves the iPhone without a deliberate action: nothing is sent over the internet.

17 About

LiftMetrics is developed by **Bertrand Cadet**, a physics and computer science teacher and weightlifter. The app was born from a training need: to see and measure what the eye cannot catch at full speed, with equipment everyone already has in their pocket.
The app's **About** page brings together:

- Support LiftMetrics;
- View the introduction again;
- User guide and privacy policy;
- Contact: liftmetrics@icloud.com;
- What the app does and its limits;
- Data (everything stays on the device);
- Scientific sources.



About

Data and privacy

- No account, no ads, no tracking.
- Videos, measurements and profiles stay on the iPhone; they only leave it if the user exports or shares them.
- Full-length videos filmed by the app can be deleted automatically after analysis; a trimmed clip of the lift is kept.

Limits

Measurements are estimates: their accuracy depends on the camera angle (side view perpendicular to the bar), plate sharpness and how the sensor is mounted. Elite reference values are digitised from published figures and are therefore approximate. The app supports training and does not replace a coach's eye.

18 Pricing

Launch period	**Everything is free, Pro included.** Users who wish can support the project with a one-off in-app tip (3 amounts, e.g. €2.99, €4.99, €6.99). A discreet invitation appears after the 5th and 20th analysis.
Free, unlimited	Snatch, clean and clean & jerk analysis: bar path, score, coaching, replay, curves; one lift at a time; history; one athlete profile; Instagram card; receiving LiftMetrics files; M5Stick sensor.
LiftMetrics Pro (planned)	**One-time purchase, no subscription** (e.g. €9.99): squat, bench press and deadlift (VBT), comparison with elite lifters, complexes and programmed sessions, multiple athletes, personal goals, load-based charts, PDF reports, CSV export, sending LiftMetrics files.
Early users	Those who installed the app during the launch get LiftMetrics Pro for free.

Indicative prices, set at App Store release. Payments go through Apple (in-app purchases); the app has no access to any banking data.

19 Ambassador proposal

LiftMetrics is an independent project, built by a coach-developer, with no investors and no ads. The goal is to grow it through the people who actually lift. An ambassador would be the voice of the app in the CrossFit community and a direct partner in shaping it.

19.1 What LiftMetrics offers

- **LiftMetrics Pro for life**, and for the ambassador's coach and training partners.
- **Early access** to every new version through TestFlight, before public release.
- **Co-design**: direct line to the developer to build the features that matter for CrossFit athletes (for example barbell cycling, touch-and-go reps, complexes typical of CrossFit programming).
- **A sensor kit** (M5StickS3, 3D-printed sleeve mount, firmware pre-installed) shipped ready to use.
- **Visibility**: the ambassador featured on the app's Instagram account and in the app's communication, with her approval of every post.
- **A personal promo code or revenue share** on Pro purchases once paid features are switched on — terms to be agreed together.

19.2 What we would ask

- Use the app in training and give honest feedback, especially during the beta.
- Occasionally share lifts analysed with LiftMetrics (bar path, velocity) on social media, at a pace that suits her.
- Allow LiftMetrics to repost that content, credited.

Next steps

1. A short call or message exchange to discuss the idea and her expectations.
2. Invitation to the TestFlight beta (iPhone required).
3. Agreement on the terms above, adjusted to suit both sides.

Contact: Bertrand Cadet — liftmetrics@icloud.com — Instagram @lift_metrics

20 Scientific references

Reference values, default goals and comparison athletes are based on the following work.

Curves and bar paths of reference athletes

[1] Japan Weightlifting Association, Japan Institute of Sports Sciences (2018). *Kinematic characteristics of the snatch techniques of the top weightlifters — 2018 IWF World Weightlifting Championships (Ashgabat)*. Scientific Report.

[2] Ang C.L., Kong P.W. (2023). Field-Based Biomechanical Assessment of the Snatch in Olympic Weightlifting Using Wearable In-Shoe Sensors and Videos — A Preliminary Report. *Sensors*, 23(3), 1171. doi:10.3390/s23031171 (CC BY 4.0).

Clean and jerk benchmarks

[3] Liu G., He J., Ye J. et al. (2024). Comparative Analysis of the Kinematic Characteristics of Lunge-Style and Squat-Style Jerk Techniques in Elite Weightlifters. *Life*, 14(9), 1086 (CC BY).

[4] Ai Q., Bi Z., Liu G. (2018). Bar Heights Needed for Successful Lifts in Men's Weightlifters. *ISBS Proceedings*, 36, 217.

[5] Núñez Othón G. et al. (2015). Evaluación técnica del clin en halterofilia femenil. *Biotecnia*, 17(E1), 28–33.

[6] Prassas S., Fulton J. A Kinematic Analysis of the Jerk Technique in Olympic Weightlifting. *ISBS Proceedings*.

Snatch reference values (default goals)

[7] Gourgoulis V. et al. (2000). Three-dimensional kinematic analysis of the snatch of elite Greek weightlifters. *Journal of Sports Sciences*, 18, 643–652.

[8] Kuhn M., Carlson K., Hoover D., Zebas C. Horizontal bar displacements of women weightlifters during the snatch. *ISBS Proceedings*.

[9] Wang C.-H., Chiu H.-T. (2008). Variability of barbell trajectory during snatch for an elite young female weightlifter. *ISBS 2008*, 644–647.

[10] Tunçel A., Harbili E., Harbili S. (2025). Comparison of snatch performance and barbell trajectory in elite weightlifters according to bodyweight categories. *Journal of Human Sport and Exercise*, 20(3), 883–894.

[11] Stone M.H. et al. (1998). Analysis of bar paths during the snatch in elite male weightlifters. *Strength and Conditioning*, 20(4), 30–38.

A Glossary

Snatch	The bar goes from the floor to locked arms overhead in one movement.
Clean	The bar goes from the floor to the shoulders.
Clean & jerk	Clean then jerk: the bar goes from the shoulders to locked arms overhead.
Complex	Several movements chained within the same set (e.g. 1 snatch + 1 clean & jerk).
First / second pull	Phases of the bar's rise, before and after the knees re-bend under the bar.
Drop under the bar	Difference between the bar's peak height and its catch height.
Bar path type	Classic classification of the bar path depending on whether it crosses the starting vertical (types 1 to 3).
1RM	Maximum load lifted for one repetition.
Mean ascent velocity	Vertical displacement of the bar divided by the duration of the ascent (mean concentric velocity, MCV).
Velocity loss	Drop in mean ascent velocity compared with the fastest rep of the set.
Sticking point	Moment of the ascent where the bar slows down before accelerating again.
VBT	Velocity-based training: using bar speed to choose loads and manage fatigue.



LiftMetrics · liftmetrics@icloud.com